Linear First-Order Equations & the Integrating Factor

ANSWER KEY

Section 1: Standard Linear Form and Integrating Factor

1 Write the differential equation $x\frac{dy}{dx} + 2y = 4x^2$ in standard form $\frac{dy}{dx} + P(x)y = Q(x)$ for $x > 0$.

a. $\frac{dy}{dx} + \frac{2}{x}y = 4x$ ✓

b. $\frac{dy}{dx} + 2y = 4x$

c. $\frac{dy}{dx} + \frac{2}{x}y = 4x^2$

d. $\frac{dy}{dx} = 4x - 2y$

Divide by the leading coefficient x: $\frac{dy}{dx} + \frac{2}{x}y = 4x$.

2 Compute the integrating factor $I(x) = e^{\int P(x)dx}$ for $\frac{dy}{dx} + \frac{2}{x}y = 4x$.

a. x^2 ✓

b. $2\ln x$

c. e^{2x}

d. x

$I(x) = e^{\int (2/x)dx} = e^{2\ln x} = e^{\ln(x^2)} = x^2$.

3 Multiply the standard ODE by $I(x) = x^2$ to solve $\frac{d}{dx}[x^2 y] = 4x^3$.

a. $y = x^2 + \frac{C}{x^2}$ ✓

b. $y = 4x^2 + C$

c. $y = x^4 + C$

d. $y = x^2 + C$

Integrate: $x^2 y = \int 4x^3 dx = x^4 + C \Rightarrow y = x^2 + \frac{C}{x^2}$.

Section 2: Solving Linear First-Order IVPs

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4 Solve the IVP $\frac{dy}{dx} + y = e^{2x}$ with $y(0) = 1$.

a. $y = \frac{1}{3}e^{2x} + \frac{2}{3}e^{-x}$ ✓

b. $y = e^{2x}$

c. $y = \frac{1}{3}e^{2x} + e^{-x}$

d. $y = \frac{1}{2}e^{2x} + \frac{1}{2}e^{-x}$

$I(x) = e^x$. $\frac{d}{dx}[e^x y] = e^{3x} \Rightarrow e^x y = \frac{1}{3}e^{3x} + C \Rightarrow y = \frac{1}{3}e^{2x} + Ce^{-x}$. Using $y(0) = 1 \Rightarrow 1 = 1/3 + C \Rightarrow C = 2/3$. Thus $y = \frac{1}{3}e^{2x} + \frac{2}{3}e^{-x}$.

5 Solve $\frac{dy}{dx} - 3y = 0$ with $y(0) = 4$.

a. $y = 4e^{3x}$ ✓

b. $y = 4e^{-3x}$

c. $y = 3e^{4x}$

d. $y = 4 + 3x$

This is a homogeneous linear equation with integrating factor e^{-3x}: $y = Ce^{3x}$. With $y(0) = 4$, $y = 4e^{3x}$.

6 What is the domain of validity for the solution to $x\frac{dy}{dx} + y = 1$ with $y(1) = 2$?

a. $(0, \infty)$ ✓

b. $(-\infty, \infty)$

c. $[1, \infty)$

d. $(-\infty, 0)$

Standard form $\frac{dy}{dx} + \frac{1}{x}y = \frac{1}{x}$ has coefficients discontinuous at $x = 0$. By the Existence and Uniqueness Theorem, the solution through $x_0 = 1 > 0$ exists uniquely on $(0, \infty)$.

Section 3: Mixing and Tank Applications

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7 A tank contains 100 L of water and 10 kg of dissolved salt. Pure water flows in at 3 L/min, and the stirred mixture drains at 3 L/min. What is the differential equation for the salt amount $A(t)$ in kg?

a. $\frac{dA}{dt} = -\frac{3}{100}A$ ✓

b. $\frac{dA}{dt} = 3 - \frac{3}{100}A$

c. $\frac{dA}{dt} = -3A$

d. $\frac{dA}{dt} = 10 - \frac{3}{100}A$

Rate in $= 0$ (pure water). Rate out $= 3 \times \frac{A}{100} = \frac{3}{100}A$. Differential equation: $\frac{dA}{dt} = -0.03A$.

8 For the tank above, what is the amount of salt remaining after t minutes?

a. $A(t) = 10e^{-0.03t}$ ✓

b. $A(t) = 10 - 0.03t$

c. $A(t) = 10e^{-3t}$

d. $A(t) = 10e^{0.03t}$

Separating variables: $A(t) = A(0)e^{-0.03t} = 10e^{-0.03t}$ kg.

9 If salt solution enters at 2 kg/L at a rate of 3 L/min, what is the inflow rate of salt in kg/min?

a. 6 kg/min ✓

b. 2 kg/min

c. 3 kg/min

d. 1.5 kg/min

Rate in $=$ concentration $\times$ flow rate $= (2$ kg/L$) \times (3$ L/min$) = 6$ kg/min.

Section 4: Transient and Steady-State Solutions

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10 In the solution $y(t) = 5 + 4e^{-2t}$, which part represents the transient response and which represents the steady-state?

a. 5 **is steady-state,** $4e^{-2t}$ **is transient** ✓

b. $4e^{-2t}$ is steady-state, 5 is transient

c. Both are steady-state

d. The entire response is transient

Because $\lim_{t\to\infty} 4e^{-2t} = 0$, that term dies out and is called transient; the permanent value 5 is the steady-state response.

11 What is the time constant τ for the decaying exponential $e^{-t/\tau} = e^{-5t}$?

a. $\tau = 0.2$ seconds ✓

b. $\tau = 5$ seconds

c. $\tau = 1$ second

d. $\tau = 2.5$ seconds

Matching $e^{-t/\tau}$ with e^{-5t} gives $\frac{1}{\tau} = 5 \Rightarrow \tau = \frac{1}{5} = 0.2$.

12 After how many time constants ($t = n\tau$) has a decaying transient decayed to less than 1% of its initial value?

a. 5 **time constants (**$e^{-5} \approx 0.0067$**)** ✓

b. 1 time constant

c. 3 time constants

d. 10 time constants

At $t = 5\tau$, $e^{-5} \approx 0.0067 < 0.01$ (less than 1% remaining), which is the standard engineering threshold for steady-state.

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